Project: Mission 01: Retail Sales Analysis with Microsoft Excel

PART 1: Data Understanding

1. Number of Datasets Provided

A total of **10 files** were provided, consisting of **9 operational datasets** and **1 data dictionary**. The datasets contain information about customers, products, sales transactions, stores, product returns, employees, suppliers and inventory.

2. Dataset Descriptions and Primary Keys

Dataset	Records	Description	Primary Key
Customers	5,000	Customer personal details and membership segments	Customer_ID
Products	500	Product names, categories, brands and prices	Product_ID
Orders	50,000	Customer orders, dates and transaction status	Order_ID
Stores	25	Store names, states and regions	Store_ID
Returns	2,500	Returned orders, reasons and refunds	Return_ID
Employees	125	Employee details and store assignments	Employee_ID
Suppliers	489	Supplier identification and contact information	Supplier_ID
Inventory	500	Available stock quantities and reorder levels	Inventory_ID
Order_Items	50,000	Products purchased, quantities, prices and discounts	Order_Item_ID

The Data Dictionary contains 41 entries describing dataset columns. It serves as reference documentation rather than a transactional table.

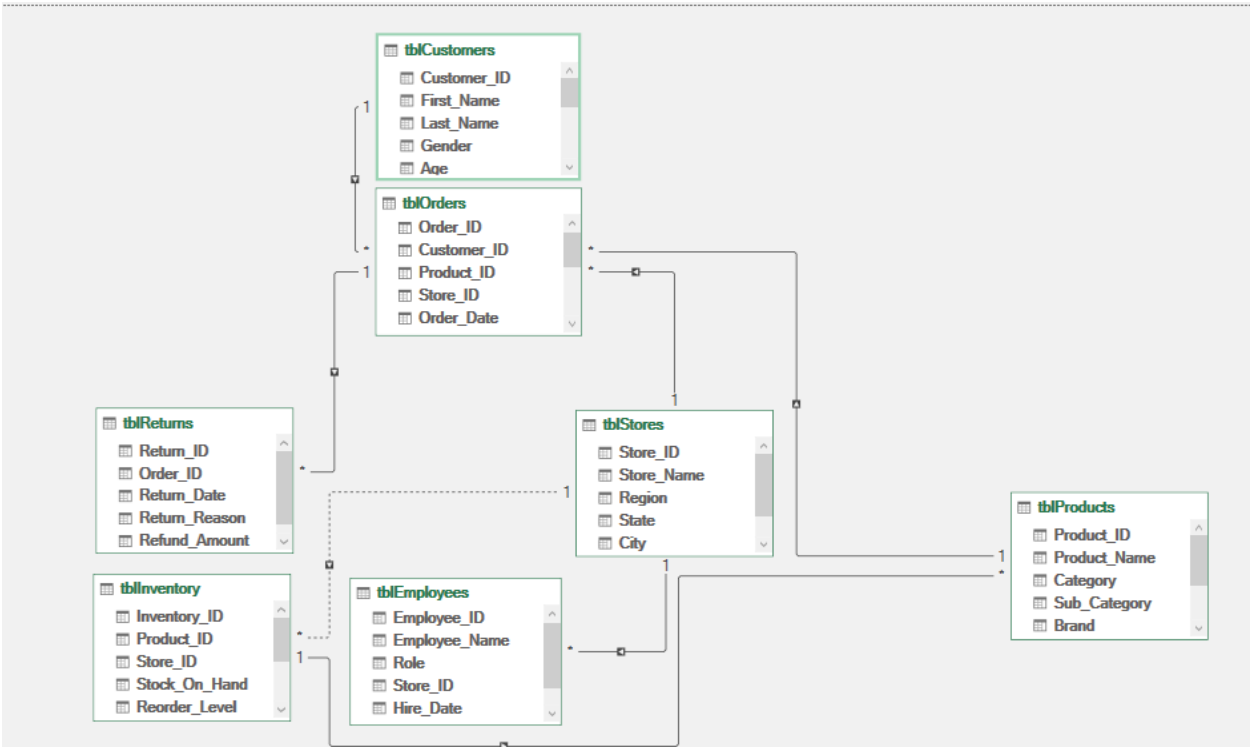
3. Identified Foreign Keys

- **Orders:** Customer_ID, Product_ID, Store_ID
- **Order_Items:** Order_ID, Product_ID
- **Returns:** Order_ID
- **Employees:** Store_ID
- **Inventory:** Product_ID, Store_ID

The Products table does not contain a Supplier_ID. However, supplier information can be linked using the product Brand and supplier name after standardizing their formats.

4. Relationships Between Tables

The relationships are primarily **one-to-many (1)**:



- Customers (1) → Orders (M)
- Products (1) → Orders (M)
- Stores (1) → Orders (M)
- Orders (1) → Returns (M)
- Products (1) → Inventory (M)
- Stores (1) → Inventory (M)
- Stores (1) → Employees (M)

Assumption:

The 'Products' table does not have Supplier_ID, but in the extract every product Brand matches a supplier name after removing the suffix "Supplies". This is a plausible mapping in this dataset. In fact, inspection found a match for all 500 product brands after removing the supplier suffix.

Note: All analyses are based on completed orders rather than pending or cancelled orders because completed orders represent transactions that have been fulfilled and provide a more reliable basis for evaluating actual sales performance, revenue, and customer purchasing activity. Excluding pending and cancelled orders helps prevent the overstatement of sales figures and ensures that the findings reflect completed transactions.

PART 8: Business Questions

1. **Which product generated the highest revenue?**

PROD0210-Beauty Item 210 generated the highest revenue, generating ₦ 7,375,596.40 in 2025

2. **Which store performed best?**

STR003 (ShopSmart Lagos) performed best in terms of sales/revenue generating ₦ 102,203,162.70 in 2025.

3. **Which category sells the most?**

Beauty was the leading category, generating revenue of ₦ 523,127,960.15 and selling 11,590 units through completed orders.

4. **Which month generated the highest revenue?**

December 2025 was the highest-revenue month, with completed-order sales of 201,038,740.75.

5. **Which customers spend the most?**

CUST03257 (Melissa Shaffer) spent the most: 2,022,739.55 across 10 completed orders.

6. **Which products are frequently returned?**

PROD0304 (Groceries Item 304) recorded the highest number of validated returned orders, with five distinct returns. However, PROD0394 (Fashion Item 394) recorded the highest validated return rate of 13.79%, representing four returned orders out of 29 completed orders.

7. **Which supplier supplies the highest number of products?**

Williams Group Supplies recorded the highest product transaction frequency, with 213 Product_ID occurrences in the Sales_Analysis dataset. However, when counting distinct products in the Products catalogue, 11 suppliers, including Williams Group Supplies, tied for the highest number, with two products each.

8. **Which stores have low inventory?**

STR017 (ShopSmart Abuja) recorded the lowest individual stock quantity, with only 10 units of PROD0479 remaining against a reorder level of 29 units. Additionally, STR014 (ShopSmart Port Harcourt) and STR006 (ShopSmart Kaduna) each recorded three inventory items at or below their reorder levels, indicating a broader need for stock replenishment.

9. **Which region performs best?**

South was the highest-revenue region, generating ₦ 724,500,644.80 from completed orders in 2025.

10. **Which product category should management invest in?**

Based on the sales analysis, ShopSmart Retail Ltd. should prioritise investment in the Beauty category, which generated the highest revenue of ₦ 523,127,960.15 and recorded the highest sales volume of 11,590 units in 2025. This performance indicates strong customer demand and suggests that further investment in the category could increase revenue. However, Beauty also recorded a relatively high product return rate, which presents a potential risk to profitability. Management should therefore invest in expanding high-performing Beauty products, maintaining adequate inventory levels, improving product quality and addressing the causes of customer

returns. Although Electronics also performs well, its higher return rate raises additional concerns. Therefore, Beauty represents the strongest opportunity for revenue growth, provided management implements measures to reduce returns and confirms that the category delivers satisfactory profit margins.

PART 9: One-Page Business Report

ShopSmart Retail Ltd. | Retail Sales Performance Analysis

Prepared for: ShopSmart Retail Ltd. Management
Prepared by: Hameedat Isa-Usman (Junior Data Analyst, AnalystLab Africa)
Reporting Period: 1 January – 31 December 2025

Overview

The analysis of 50,000 orders, including 16,529 completed transactions, recorded total completed-order revenue of ₦2,246,837,830.50. Sales were calculated using the tutor's percentage-discount method in Microsoft Excel.

Three key findings

1. Strong category demand. Beauty led all categories with ₦523,127,960.15 in revenue and 11,590 units sold.

2. Uneven sales performance. The South region generated ₦724,500,644.80, STR003 (ShopSmart Lagos) led individual stores with ₦102,203,162.70, and December was the strongest month at ₦201,038,740.75.

3. Operational concerns. At STR017 (ShopSmart Abuja), product PROD0479 had only 10 units against a reorder level of 29. PROD0394 (Fashion Item 394) had a 13.79% validated return rate (4 of 29 completed orders).

Three recommendations

1. Invest selectively in Beauty. Prioritise high-demand products and assess profit margins before expanding the range.

2. Strengthen stock controls. Set reorder alerts, address the STR017 shortage, and review other stores with several low-stock items.

3. Reduce avoidable returns. Investigate product defects, descriptions and return reasons; monitor return rates by product and category.

Risk and opportunity

Risk: Stockouts and returns could erode sales and customer trust. Inconsistent return dates and order statuses also limit the reliability of return metrics.

Opportunity: Targeted Beauty promotions and stock allocation in stronger-performing regions could increase sales without indiscriminate inventory expansion.

Note: *The ₦ designation follows the Nigerian project scenario.*